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# Engineering Paradigms for Sheared-Flow-Stabilized Z-Pinch Fusion Energy

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**Abstract** — *The sheared-flow-stabilized Z-pinch concept is on a path to commercialization at Zap Energy. Recent experiments on the Fusion Z-pinch Experiment (FuZE) device corroborate expected plasma stability and thermonuclear fusion reaction rates. Experimental campaigns are underway to increase the pinch current, the stable plasma duration, and the DD fusion neutron production. The next-generation device FuZE-Q is currently undergoing commissioning and will begin operation at current levels where scientific breakeven-equivalent conditions are expected in the near future. The Z-pinch configuration offers the promise of a compact fusion device owing to its simple geometry, unity beta, and absence of external magnetic field coils.*

*In addition to a robust experimental program pushing plasma performance toward breakeven conditions, Zap Energy has parallel programs developing power handling systems suitable for future power plants. Technologies under development include high-average-power repetitive pulsed power, high-duty-cycle cathodes, and liquid-metal wall systems. High-level features of the conceptual power plant core design are elaborated and compared with other approaches to fusion energy.*

**Keywords** — Z-pinch, alternative fusion concept, deuterium-tritium, compact fusion reactor, fusion experimental reactor design.

**Note** — *Some figures may be in color only in the electronic version.*

## I. INTRODUCTION

Interest in controlled nuclear fusion as a power source is growing rapidly due to the acute need for large-scale sources of carbon-free electrical power and process heat to mitigate climate change. Sixty years of past plasma confinement research have produced steady increases in performance as measured by the Lawson criterion.<sup>[1]</sup> Recent

advances toward scientific breakeven, the point at which the fusing plasma produces more energy from fusion reactions than the energy used to ionize and heat it, have spurred the development of a nascent private fusion industry.<sup>[2]</sup> The achievement of net energy production from a confined fusing plasma is a necessary but insufficient condition for the creation of a practical and economically attractive power plant. Issues surrounding confinement and heating system complexity, fuel cycle, longevity and maintainability of components, total radioactive inventory, and operational overhead will drive the cost of future fusion energy.

Lidsky clearly showed that the largest impediments to an economic fusion power plant group into two categories.<sup>[3]</sup> The first category revolves around the plasma confinement device's large size, high technological complexity, and low power density compared with alternatives like fission-based nuclear energy. The second category

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focuses on the neutron intensity of the deuterium-tritium (DT) fuel cycle and the associated issues of radiation damage in materials, induced radioactivity, and shielding. Both sets of issues drive up the cost of the prototypic magnetic confinement (tokamaks and mirror machines) and inertial confinement (both laser- and particle beam-driven) DT-fueled systems.

Lidsky advocated, as have others before and since, for focusing on lower-neutron-intensity fuel cycles, particularly deuterium with  $^3\text{He}$  ( $\text{D-}^3\text{He}$ ) and proton with  $^{11}\text{B}$  ( $\text{p-}^{11}\text{B}$ ), to mitigate these factors and prevent them from making future fusion-generated power economically unattractive.<sup>[3]</sup> However, the risks of pursuing this strategy are high and driven by the temperatures and confinement times necessary to burn these fuels.<sup>[1]</sup> While  $\sim 10\text{-keV}$  thermal temperature is generally sufficient for a DT reactor,  $\text{D-}^3\text{He}$  requires  $\sim 70\text{ keV}$ <sup>[4]</sup> and  $\text{p-}^{11}\text{B}$   $\sim 100\text{ keV}$ .<sup>[1]</sup> Given that six decades of research are just now bringing breakeven thermal DT plasma conditions within grasp, it is unclear when the order of magnitude improvements in the thermal plasma temperatures and confinement necessary for  $\text{D-He}^3$  and  $\text{p-B}^{11}$  will be available. A non-Maxwellian system based on high energy beam injection into a colder ion target can successfully generate energy from these fuels in principle, but the very high requirements on electron temperature in such a system exceed what has been achieved experimentally by a wide margin.<sup>[5–7]</sup>

The sheared-flow-stabilized (SFS) Z-pinch approach to fusion energy<sup>[8]</sup> aims to improve the economic viability of fusion power by creating a system that can use the DT fuel cycle as advantageously as possible. Compared with typical magnetic and inertial confinement approaches, the SFS Z pinch is intrinsically small in physical size, low in technical complexity, and high in power density. Sections II and III of this paper elaborate on the history of Z-pinch fusion research and the short ( $\sim 0.5\text{ m}$ ), thin ( $\sim 0.15\text{ mm}$ ), and high density ( $\sim 10^{26}\text{ m}^{-3}$ ) SFS Z-pinch plasmas planned for prototypical power plants based on the technology. Section IV describes how the SFS Z pinch is directly and efficiently driven by an electrical power supply without the need for external magnets, auxiliary heating, or inefficient conversion of the drive energy to other intermediate forms. Section V discusses how the linear geometry of the SFS Z-pinch enables relatively straightforward engineering implementation of a liquid-metal first wall and breeding blanket compared with other fusion approaches. Operating a DT fusion plasma within a liquid-metal first wall and blanket has the potential to greatly reduce the engineering problems associated with first-wall heat flux, neutron radiation damage, and activation. Finally, Sec. VI provides a summary of this paper.

## II. Z-PINCH FUSION

Z-pinch fusion relies on the self-compression that occurs in a plasma column due to the passage of a strong unidirectional current and has a long history in controlled thermonuclear fusion research. Early promising results on the ZETA device in the late 1950s briefly made Z-pinches the front runner in fusion energy research.<sup>[9]</sup> However, progress in Z-pinch performance was short-lived due to the rampant plasma instabilities endemic to a basic Z-pinch. By the 1960s, research on Z-pinches as systems for fusion energy was largely abandoned in favor of other approaches. This situation began to change in the 1990s with the proposal to stabilize Z-pinch plasmas via sheared flow.<sup>[10]</sup>

The SFS Z-pinch concept,<sup>[8]</sup> originally developed at the University of Washington with Lawrence Livermore National Laboratory collaborators, is now on a path to commercialization at Zap Energy. The Fusion Z-pinch Experiment (FuZE) employs high power-handling electrodes, flexible gas injection, and independently switched capacitor bank modules to tailor the discharge current and gas distribution to establish stabilizing sheared flow and pinch current (Fig. 1).<sup>[11]</sup> Recent experiments corroborate the expected DD fusion neutron production rates sustained for  $\sim 10\text{ }\mu\text{s}$ , which is thousands of times the magnetohydrodynamic instability growth time typical of an unstabilized Z-pinch with similar parameters.<sup>[12]</sup> The measured neutron spectra are also consistent with thermonuclear production.<sup>[13]</sup>

Experimental campaigns are underway to increase the pinch current, stability duration, and DD fusion neutron production. The next-generation device FuZE-Q is currently undergoing commissioning and will soon begin operation at currents levels where scientific breakeven-equivalent conditions are expected. The Z-pinch configuration offers the promise of a compact fusion device

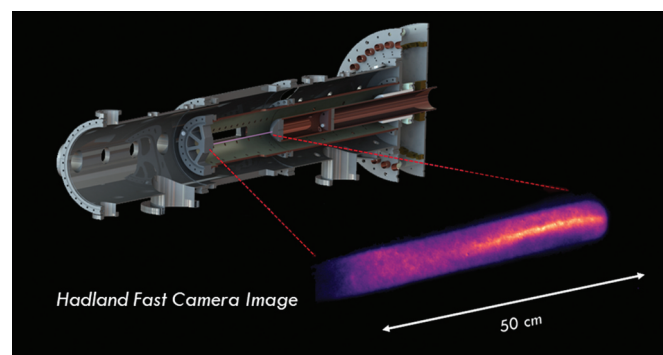


Fig. 1. False color, visible wavelength image of a SFS Z-pinch plasma (lower right) shown with the approximate location the plasma occupies within the FuZE device.

owing to its simple geometry, unity beta, and absence of external magnetic field coils.

### III. SFS Z-PINCH DT PLANT PROTOTYPICAL PARAMETERS

The SFS Z pinch is an inherently compact form of plasma confinement that enables similarly compact power plant designs. Table I lists the approximate plasma parameters at each step between the FuZE and a nominal SFS Z-pinch power plant core.<sup>[14]</sup> In all cases, the plasma length remains 0.5 m long. A single plasma temperature  $T$  is given since equal ion temperature  $T_i$  and electron temperature  $T_e$  are assumed in SFS Z-pinch plasmas.<sup>[8]</sup> That assumption was validated on FuZE with independent  $T_i$ <sup>[14]</sup> and  $T_e$ <sup>[15]</sup> measurements. As shown in Table I, Z-pinch plasma parameters elevate rapidly with increasing pinch current. Delivering pinch currents of  $\sim 1$  MA is well within the technical state of the art in the field of pulsed power and numerical simulations suggest that sheared-flow stabilization is robust, but the question remains if sheared flows will continue to be effective at stabilizing laboratory Z pinches with higher fusion performance and longer pulse durations.<sup>[8]</sup> The FuZE-Q device was designed and built to explore this question experimentally.

The current conceptual core design uses a vertically oriented plasma operated in pulsed mode (Fig. 2). A tank of liquid metal serving the dual functions of heat exchange and tritium breeding medium forms the main body of the conceptual core. The liquid is pumped from the sides of the tank up and over a weir wall to form a central cavity surrounded by flowing liquid metal. The SFS Z-pinch formation system sits on top of the tank and repeatedly fires pulses of plasma that assemble and generate fusion reactions within the liquid-metal-lined cavity. Each plasma pulse will produce a nominal yield of 19 MJ

of fusion energy, as described in Table I. The pulse rate is variable to allow for controllable output power and load-following with a nominal maximum thermal power of 200 MW.

A power generation plant could have several SFS Z-pinch fusion cores to increase maximum power output and reduce costs by sharing maintenance and the tritium handling infrastructure among multiple cores. The core's size of approximately 3 m in diameter is set by the required thickness of the liquid blanket, which is composed of lithium and lead in the current conceptual design. The following sections detail the advantages of this approach to fusion energy.

### IV. COMPLEXITY AND EFFICIENCY OF PLASMA CONFINEMENT AND HEATING SYSTEMS

A clear advantage of the SFS Z-pinch approach to fusion energy is the relative simplicity of the device. Focusing just on the major systems required to confine, heat, and burn the thermonuclear plasma, a typical tokamak requires superconducting magnetic field coils, a cryogenic plant, a vacuum system, and electrical power supplies. A SFS Z-pinch device only requires a vacuum system and electrical power supplies. Dispensing with the superconducting magnets and cryogenic plant is already a substantial reduction in technical complexity. Furthermore, nine types of electrical power supplies are necessary for typical tokamak fusion devices.<sup>[16]</sup> Four of these power supply categories are dedicated to the magnetic coils and auxiliary heating systems, none of which are necessary for a SFS Z-pinch system. The remaining five categories dealing with driving plasma current and facility load are common to both approaches to fusion. Therefore, the engineering of a SFS Z-pinch-based power plant begins at a substantially lower

TABLE I  
SFS Z-pinch Approximate Plasma Parameters at Each Development Step

| Parameter        | Symbol                   | FuZE                   | Scientific Breakeven | Engineering Breakeven | Power Plant (Each Core) |
|------------------|--------------------------|------------------------|----------------------|-----------------------|-------------------------|
| Pinch current    | $I_{pinch}$ (MA)         | 0.25 to 0.3            | 0.6 to 0.7           | 1.0                   | 1.2 to 1.5              |
| Pinch radius     | $a$ (mm)                 | 1 to 3                 | 0.25                 | 0.2                   | 0.15                    |
| Pinch length     | $l_p$ (m)                | 0.5                    | 0.5                  | 0.5                   | 0.5                     |
| Electron density | $n_e$ (m <sup>-3</sup> ) | $10^{23}$ to $10^{24}$ | $5 \times 10^{25}$   | $10^{26}$             | $1.5 \times 10^{26}$    |
| Temperature      | $T$ (keV)                | 1 to 2                 | 7                    | 10 to 15              | 15 to 30                |
| Plasma lifetime  | $\tau_p$ ( $\mu$ s)      | 20 to 40               | 100                  | 150                   | 200                     |
| Fusion energy    | $E_{fus}$ (MJ)           | $10^{-6}$              | 0.07                 | 1                     | 19                      |

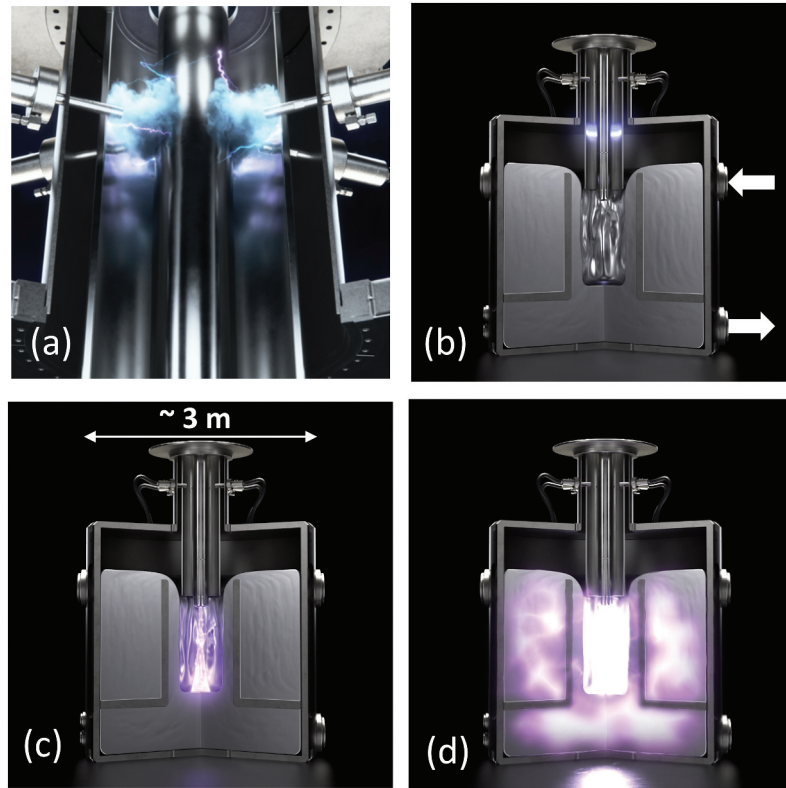


Fig. 2. Artist's conception of a SFS Z-pinch fusion power core. (a) DT gas injection and ionization occur near the top of the device. (b) The plasma travels down the coaxial accelerator region toward a cavity formed by pumping liquid metal up and over a weir wall within a larger tank. White arrows denote the liquid-metal inlet and outlet. (c) SFS Z-pinch plasma column assembles on axis and fuses. (d) Fusion neutrons are captured in the liquid-metal blanket.

level of interacting systems and complexity than a prototypical tokamak (Fig. 3). Tokamaks have had a longer development history than SFS Z-pinch devices and are currently much closer to breakeven performance as measured by the Lawson criterion.<sup>[1]</sup> The degree to which SFS Z-pinch plasma confinement and heating systems will increase in complexity as their performance increases remains to be seen.

The SFS Z-pinch is an explicitly pulsed device. A power plant based on this technology would operate in a manner analogous to an internal combustion engine. Each individual firing of the stabilized Z-pinch would be a discrete event that produces ~20 MJ of fusion energy (Table I). The rate of firing, much like the revolutions per minute of an internal combustion engine, may be varied to adjust total power output as needed. This operational paradigm opens up possibilities for effective load-following to better support a grid powered with a significant fraction of variable wind and solar sources.

Considerations of the recirculating power fraction and capacity factor, which are both vital to the economic viability of fusion energy, highlights the advantages of

the SFS Z-pinch in pulsed-mode operation. Recirculating power refers to the amount of electricity generated from fusion power that must be fed back into the plant to maintain its operation. All fusion power systems have substantially higher recirculating power needs than legacy fossil fuel or fission power plants due to the need for plasma confinement and heating to maintain the fusion reaction. The capacity factor is the time-averaged power produced by the plant divided by the nominal plant design power. Early generations of fusion power plant designs based on steady-state tokamak concepts are expected to barely produce net power based on estimates of recirculating power between 0.4 to 0.6, capacity factors of 0.4 to 0.8, and the commensurate poor overall plant efficiency.<sup>[17]</sup>

One key factor in this analysis is that tokamaks are designed to operate at a fixed magnetic field and plasma current.<sup>[17]</sup> Therefore, the current drive power, which is a dominant part of the recirculating power, does not scale with the output power. This is not the case in a pulsed-mode SFS Z pinch where the power output is expected to scale with the input current and power,<sup>[8]</sup> improving



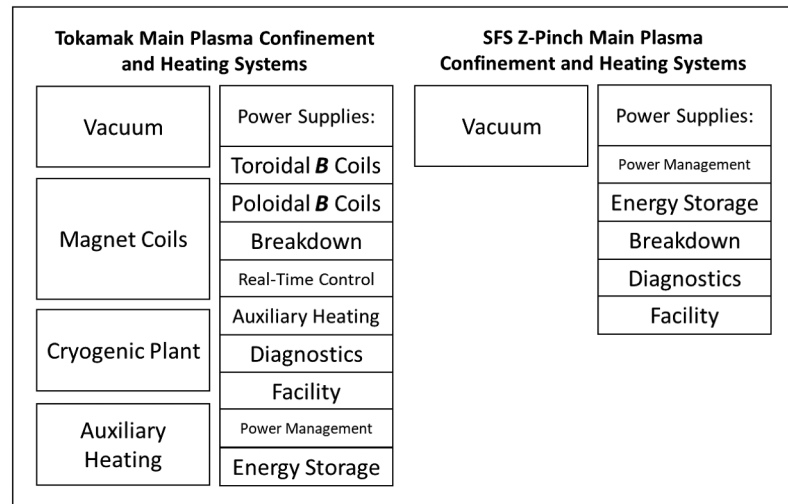


Fig. 3. Graphical representation of the relative complexity of plasma confinement and heating systems in tokamaks and SFS Z-pinch devices. Power supply categorization for tokamaks is adapted from Lampasi and Minucci<sup>[16]</sup> and mapped approximately to SFS Z-pinch systems for comparison purposes.

overall plant efficiency when variable output is required. While other approaches to fusion, like inertial confinement fusion (ICF), share some advantages of pulsed-mode operation, they suffer from extreme inefficiencies in their approaches to plasma heating. Projections based on current and near-term laser technology suggest that electrical-to-optical energy efficiency in a future National Ignition Facility (NIF)-derived laser ICF system could reach 18%.<sup>[18]</sup> This is a much higher efficiency than the current state of the art at the NIF and would require heavily optimizing the 17 loss mechanisms analyzed by Bayramian et al. in the lengthy conversion chain between alternating current (AC) power and laser light.<sup>[18]</sup>

The electrical energy necessary to confine and heat the SFS Z-pinch is driven directly from the power supply to the plasma without highly inefficient intermediate conversion to other forms (e.g., neutral particle beams, lasers, or radio-frequency electromagnetic waves). The SFS Z-pinch driving power supply consists mainly of AC to direct current (DC) rectification followed by a pulsed-power modulator. The AC-DC rectification efficiency of ~90% has been documented at the long-running 12.5-MW High Intensity Proton Accelerator (HIPA) cyclotron-based proton accelerator facility.<sup>[19]</sup> Pulsed-power modulator units based on solid-state thyristor switch stacks with current and voltage ratings relevant to future SFS Z-pinch power plants have demonstrated 80% efficiency via calorimetric measurement when operating for hours at five pulses per second.<sup>[20]</sup> Taking these numbers as a basis and rounding down, the total efficiency of the electrical drive system for the SFS Z pinch from the “wall plug” to the plasma cathode is expected to

be approximately 70%. The repetitive pulsed-power program at Zap Energy is currently building subscale test stands to evaluate and develop power supply technology for future power plants.

## V. HIGH COVERAGE LIQUID-METAL FIRST WALL AND BLANKET

In addition to confining, heating, and fusing the fuel, any practical fusion power plant based on the DT fuel cycle must economically turn the high-energy neutrons produced into electricity and more tritium. The preferred engineering approach to this requirement is minimizing the solid structure near the fusing plasma and designing the first wall and neutron-absorbing tritium breeding blanket to consist primarily of a flowing, lithium-bearing liquid coolant. Neutrons absorbed in a liquid are prevented from striking and embrittling solid parts of the core structure. A flowing first wall can also tolerate higher heat fluxes than a solid first wall. Finally, a combined liquid coolant and breeding medium can be pumped through the core to facilitate the extraction of both heat and tritium.

Much work has been done on liquid surfaces for plasma-facing components in tokamaks, but it has been largely focused on thin layers for mitigating heat flux and impurity recycling from the vacuum chamber wall.<sup>[21]</sup> Lithium is preferred in this application since it is the lightest nongaseous element, which makes it a relatively benign plasma impurity and a good chemical getter of other common heavier impurities. However, the toroidal geometry and high magnetic

fields of tokamaks make the use of any flowing liquid metal difficult due to magneto-hydrodynamic and gravitational forces and precludes any obvious technical path to a thick liquid first wall.

Here we use the term thick liquid first wall to denote a width of liquid material, likely on the order of a meter or more, sufficient to significantly reduce the neutron flux and the commensurate neutron damage rate impinging on the main solid structural components of the fusion device. While there is little prospect of reducing neutron damage to the plasma-facing solid mechanical structures of tokamaks using a thick liquid first wall, tritium breeding in nonconducting molten salts behind the first-wall system is an active area of development.<sup>[22]</sup>

Thick liquid first walls are much more feasible in inertial and magneto-inertial fusion concepts where there are either no strong magnetic fields in the system or the strong magnetic fields do not extend outside the plasma itself. The absence of magnetic fields, and hence magneto-hydrodynamic forces, in the first-wall and blanket regions significantly reduces the difficulty of pumping liquid metal through these volumes. The High-Yield Lithium-Injection Fusion-Energy (HYLIFE) design study envisioned using 175 vertical liquid lithium jets surrounding the point where ICF fuel pellets are repeatedly exploded to protect the first structural wall from X-rays, alpha particles, debris, and 90% of the neutron flux.<sup>[23]</sup>

The HYLIFE-II considered the same design with a molten salt composed of fluorine, lithium, and beryllium ( $\text{Li}_2\text{BeF}_4$ ) called FLiBe, replacing pure molten lithium.<sup>[24]</sup> This avoids the fire hazard of pure lithium, but also complicates the chemical environment within the core and introduces beryllium along with its associated toxicity hazards. Similar approaches using liquid jets have been studied for inertial fusion energy driven by unstabilized Z pinches.<sup>[25]</sup>

Common to all these approaches is the large size of the power plant overall and the magnitude of the individual pulses of fusion energy (Table II). SFS Z pinches are expected to operate effectively at a comparatively small individual pulse fusion yield. The lower individual pulse

yield relaxes the requirements around protecting the solid structures near the plasma from the pressure shock of the fusion burst, in addition to the average heat loads and cumulative neutron damage, leading to substantially smaller individual core volumes.

Proposals for alternative concepts with thick liquid walls include spheromaks<sup>[26]</sup> and spherical tokamaks.<sup>[27]</sup> While some of these are at a smaller scale than the ICF plant proposals, they typically involve spinning the molten wall/blanket liquid to generate centrifugal forces that push it against the walls in order to maintain openings in the fluid at both the top and bottom, which is required for access to the plasma in these approaches.<sup>[26]</sup> Spinning tons of molten metal or salt fast enough to hold it on a vertical wall against the force of gravity adds significant engineering complexity to the design. Some magnetized target fusion concepts involve rapidly driving large masses of molten metal to act as a compression liner for a spherical tokamak plasma.<sup>[27]</sup> This approach adds issues of timing and high repetitive mechanical stress to the complexities of high-speed fluid flow.

The SFS Z pinch requires neither externally driven compression from the liquid wall nor access to the plasma from the bottom of the system through the blanket. Therefore, a fusion power core based on this plasma confinement technology can implement a thick liquid wall and blanket using a simple tank with internal baffles forming a weir wall. Zap Energy's current working point design uses the eutectic mixture of lithium and lead LiPb (17% lithium and 83% lead), which is pumped into a volume and allowed to cascade over the rim of the weir wall under gravity to form the first wall and a cavity for the SFS Z-pinch plasma.<sup>[14]</sup> The pinch current terminates in the LiPb liquid at the bottom of the cavity. The LiPb has approximately the same resistivity at 600 K as stainless steel does at 300 K.<sup>[28]</sup>

At plant-relevant pinch currents (Table I), the fusion  $Q$  ( $P_{\text{fusion}}/P_{\text{input}}$ ) is greater than 10 and ohmic heating of the LiPb is a small fraction of the fusion power heating.<sup>[14]</sup> With the exception of resistive losses in the

TABLE II  
Nominal Power, Shot Yield, and Size of Selected Pulsed Fusion Concepts

| Proposed System           | Single-Core Thermal Power | Individual Pulse Fusion Yield | Core Volume        |
|---------------------------|---------------------------|-------------------------------|--------------------|
| HYLIFE <sup>[23]</sup>    | 2.7 GW                    | 1.8 GJ                        | 630 m <sup>3</sup> |
| HYLIFE-II <sup>[24]</sup> | 2.5 GW                    | 350 MJ                        | 340 m <sup>3</sup> |
| Z-IFE <sup>[25]</sup>     | 2 to 2.4 GW               | 3 to 30 GJ                    | 700 m <sup>3</sup> |
| SFS Z pinch               | 200 MW                    | 19 MJ                         | 25 m <sup>3</sup>  |

cathode structures and fusion-generated particles hitting its relatively small volume, virtually all the electrical drive power input  $P_{input}$  to the core and the fusion power  $P_{fusion}$  generated in DT reactions is deposited in the liquid-metal wall and blanket. The bulk of  $P_{input}$  becomes bremsstrahlung radiation in the SFS Z-pinch plasma, which deposits in a thin layer of the flowing liquid first wall of the pinch cavity.  $P_{fusion}$  is emitted largely in the form of 14.1-MeV neutrons that penetrate the bulk of the liquid blanket as they deposit their energy and 3.5-MeV  $^4\text{He}$  nuclei that deposit their energy in a thin layer of the first wall. The heated liquid LiPb flowing down the weir wall collects at the bottom of the tank where it is pumped out, passed through a heat exchanger and tritium extraction stage, and recirculated into the core.

Liquid LiPb has the advantages of combining the tritium breeding lithium and a neutron multiplier, via the  $\text{Pb}(n, 2n)$  reaction, in the same material and greatly reducing chemical reactivity with air and water compared to pure lithium.<sup>[28]</sup> The disadvantages of LiPb include its high density, which increases pumping power requirements, and the production of activation products like  $^{210}\text{Po}$  and  $^{203}\text{Hg}$ , though the latter can be mitigated by controlling the isotope mix in the LiPb eutectic.<sup>[29]</sup> Initial calculations of the tritium breeding ratio in a SFS Z-pinch core with a LiPb wall and blanket returned a value of approximately 1.1.<sup>[14]</sup> Zap Energy has an active materials program working to advance our thick-liquid-wall core design.

While liquid LiPb can protect most of the solid structure of a SFS Z-pinch power core from high heat flux and reduce neutron damage, the plasma cathode is an exception. The cathode carries and shapes the megaampere-scale currents necessary to form and briefly sustain each SFS Z-pinch plasma pulse.<sup>[11]</sup> This function requires a solid cathode structure that is in direct contact with the plasma, and therefore directly exposed to plasma erosion and the high neutron flux emanating from the fusing plasma. However, the cathode represents a small volume and mass of material compared to the rest of the plant core and its simple geometry facilitates straightforward remote removal and replacement.

There are decades of successful plasma arc cathode operation experience in the extreme, albeit nonnuclear, conditions of commercial arc smelting furnaces at powers up to 60 MW.<sup>[30,31]</sup> This provides confidence that practical engineering solutions can be found for SFS Z-pinch power plant cathode design. To this end, Zap Energy is working on several damage-mitigation techniques to increase cathode longevity and reduce the maintenance interval in a future power plant. Notably, the cathode structure also subtends

a relatively small solid angle as viewed from the plasma and minimally interferes with the tritium breeding process.

## VI. SUMMARY

The demonstration of high-gain plasma performance in the SFS Z pinch will open the pathway to implementing simple and flexible DT fusion power core designs. Compact plasma geometry and the absence of external magnets allows for the straightforward operation of a liquid lithium–lead first wall and blanket to address the high first-wall heat flux and DT neutron damage issues. Plasma confinement and heating electrical efficiency, as well as pulsed operation variable output characteristics, are both attractive for end-product economics compared with other fusion approaches. Subscale testing of liquid-metal wall concepts, high average power supplies, and durable cathode designs are underway to lay foundations for future SFS Z-pinch power plants.

## Disclosure Statement

No potential conflict of interest was reported by the authors.

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